

ISyE 6525 High Dimensional Data Analytics - HW2

Module 2: Image Analysis

Fall 2026

Submission. Submit *one PDF report per question*, and for every question that requires code, *one code file for that question*. You may use MATLAB, R or Python. Total: **100 points**.

Question 1.

(20 points)

Image segmentation: Otsu's method. Suppose you have an $m \times n$ gray image, with x_{ij} the intensity of pixel (i, j) , $i = 1, \dots, m$ and $j = 1, \dots, n$. The total variance of the image is

$$\sigma_T^2 = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n (x_{ij} - \mu_T)^2,$$

where μ_T is the average intensity of the image.

Now consider thresholding the histogram of the image in order to get a black-and-white image. Let t be the threshold. The goal of Otsu's method is to find an optimal value of t by minimizing the intra-class (within-class) variance

$$\sigma_w^2(t) = \omega_1(t)\sigma_1^2(t) + \omega_2(t)\sigma_2^2(t),$$

where $\omega_i(t)$ is the proportion of pixels in class i and $\sigma_i^2(t)$ is the variance of class i , $i = 1, 2$. It is easy to see that minimizing the intra-class variance is equivalent to maximizing the inter-class (between-class) variance $\sigma_b^2(t) = \sigma_T^2 - \sigma_w^2(t)$.

- (a) **(15 pts)** Show that

$$\sigma_b^2(t) = \omega_1(t)\omega_2(t)(\mu_1(t) - \mu_2(t))^2,$$

where $\mu_i(t)$ is the average intensity of class i , $i = 1, 2$.

- (b) **(5 pts)** Using the result of (a), explain why $\sigma_b^2(t)$ is exactly zero at $t = 0$ and at $t = 255$, and describe the two competing effects that determine where the maximum is attained.

Question 2.

(25 points)

Intensity transformations. Read `Flower.jpg`. All three computational questions on this assignment use that image.

- (a) **(3 pts)** Read and display the image. Report the size of the original image, then resize it to a third of its size and report the new size.
- (b) **(3 pts)** Convert the original image to a gray image and to a black-and-white image using a level of 0.45. Display both.
- (c) **(4 pts)** Display the histogram of the gray image. What can you say about the image from this histogram? Report the contrast of the gray image.

- (d) **(15 pts)** Apply the following transformations to the gray image. For each one, report the resulting image, the resulting histogram, and one sentence on the purpose of the transformation.
1. Linear transformation $g = af + b$ with $a = 1.4$, $b = -20$.
 2. Log transformation $g = c \log(1 + f)$ with $c = 40$.
 3. Power-law transformation $g = cf^\gamma$ with $c = 0.1$ and $\gamma = 1.4$.
 4. Thresholding at 120.
 5. Histogram shift: lower bound 25, upper bound 225, shift of +50.
 6. Histogram stretch with $\lambda = 235$. Report the contrast before and after.

Question 3.**(30 points)****Convolution and denoising.**

- (a) **(10 pts)** Implement your own filtering function `my_imfilter(image, kernel)` that behaves like the built-in `imfilter`. Requirements:
- Input: image, filter. Output: filtered image of the *same* size as the input.
 - It must work for colour images (filter each channel independently).
 - It must work for any kernel whose height and width are odd, and it should raise an error otherwise.
 - Pad the input image with zeros, so that the output resolution matches the input.
 - You may **not** call `imfilter`, `conv2`, `filter2`, `convolve2d` or any equivalent library routine. Loop over the pixels and do the computation yourself. It may be slow.
- (b) **(6 pts)** Convert `Flower.jpg` to grayscale and build two noisy images:
- J_1 : add zero-mean Gaussian white noise with variance 0.01 (on the $[0, 1]$ scale).
 - J_2 : add salt-and-pepper noise affecting approximately 5% of the pixels (half set to 0, half to 255).

Display both and report their PSNR with respect to the clean image,

$$\text{PSNR} = 10 \log_{10}(255^2 / \text{MSE}).$$

- (c) **(10 pts)** Apply the following five 3×3 filters to J_1 and to J_2 and display the results:
1. Arithmetic mean filter, $h = \frac{1}{9} \mathbf{1}_{3 \times 3}$.
 2. Gaussian filter $h_g(x, y) \propto \exp(-(x^2 + y^2)/(2\sigma^2))$ with $\sigma = 1$, normalized to sum to one. Report the kernel.
 3. Median filter: sort the intensities in the 3×3 neighbourhood of the reference pixel and replace the pixel by the median.
 4. Minimum filter, $\hat{f}(x, y) = \min_{(r,c) \in S_{xy}} g(r, c)$.
 5. Maximum filter, $\hat{f}(x, y) = \max_{(r,c) \in S_{xy}} g(r, c)$.

The two linear filters must go through your `my_imfilter`.

- (d) **(4 pts)** Report the MSE and PSNR of all ten filtered images in one table. Which filter do you recommend for Gaussian noise, and which for salt-and-pepper noise? Explain why in two or three sentences.

Question 4.**(25 points)**

Edge detection and segmentation.

- (a) **(8 pts)** Compute $\nabla^2 G$ for

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \quad \text{Hint: } \nabla^2 G = \frac{\partial^2 G}{\partial x^2} + \frac{\partial^2 G}{\partial y^2}.$$

Show your work. Then use your expression to construct a 5×5 Laplacian-of-Gaussian kernel with $\sigma = 1$, sampled on the integer grid $x, y \in \{-2, -1, 0, 1, 2\}$. Report the kernel numerically, explain why it should be shifted so that its entries sum to zero, convolve it with the gray image and display the result.

- (b) **(7 pts)** Compute the first partial derivatives of the gray image in the x and y directions using the Sobel operator, and form the gradient magnitude $|\nabla f| = \sqrt{f_x^2 + f_y^2}$. Repeat with the Prewitt operator. For each operator, display the thresholded edge maps for at least four cutoff values. What happens as the cutoff increases? Which operator gives the larger response, and why?
- (c) **(6 pts)** Use the k -means clustering algorithm to segment the *colour* image into 2, 3, 4 and 5 clusters. Display the four segmented images and report the cluster centres.
- (d) **(4 pts)** Use Otsu's method to segment the gray image into 2, 3, 4 and 5 levels. Report the threshold values obtained in each case and display the quantized images.